

TOPIC 7: COMPUTER BOOTING AND SHUTDOWN PROCESSES

1. Meaning of Booting

Booting is the process of starting a computer and loading the operating system into memory so that the computer is ready for use. It is also called start-up or boot process.

2. Steps in Booting (Cold Boot)

1. Power on the computer (press the power button).
2. Power-On Self-Test (POST) – the computer checks hardware (memory, keyboard, etc.).
3. BIOS/UEFI loads and looks for the boot device (usually the hard disk).
4. The boot loader finds and starts the operating system.
5. The operating system loads into RAM and displays the login screen or desktop.
6. User logs in (if required) and the computer is ready for use.

3. Types of Booting

- **Cold boot** – starting from a completely powered-off state.
- **Warm boot** – restarting the computer without turning the power off completely (Restart).

4. Steps in Shutting Down

1. Save all open work and close programs.
2. Click the Start button (or equivalent).
3. Select Shut down / Power off.
4. Wait until the computer powers off completely before switching off any extension or surge protector.

Avoid forcing shutdown by holding the power button except in emergencies, as this can cause data loss or file corruption.

5. Safe Handling of Devices

- Always shut down properly before unplugging.
- Do not move the computer roughly while it is on (especially HDD).
- Keep drinks and food away from the computer.
- Use a stable power supply or UPS where possible.

6. Summary

Booting loads the operating system so the computer can work. Shutting down should be done properly through the operating system menu to protect data and hardware.

End of Topic 7